Step 1: Read and Show Image

img = imread('peppers.png');
imshow(img);

- Loads the image (`peppers.png`)
- Displays the original image

🔹 Step 2: Convert to Grayscale

gray_img = rgb2gray(img);

- Converts color image → black & white (grayscale)
- Makes processing easier

🔹 Step 3: Remove Noise

filtered_img = imgaussfilt(gray_img, 2);

- Applies **Gaussian filter**
- Smooths the image and removes noise (small unwanted dots)

🔹 Step 4: Convert to Binary Image

threshold = graythresh(filtered_img);
binary_img = imbinarize(filtered_img, threshold);

- Uses **Otsu's method** to find best threshold
- Converts image into **black and white**

🔹 Step 5: Clean Image (Morphology)

cleaned = imopen(binary_img, se);
cleaned = imclose(cleaned, se);

- **Opening** → removes small noise
- **Closing** → fills small holes
- Result: cleaner object shapes

🔹 Step 6: Detect Objects

```
labeled = bwlabel(cleaned);
stats = regionprops(labeled, 'Area', 'BoundingBox', 'Centroid');
```

- Labels each object
- Finds:
  - **Area** (size)
  - **Bounding box** (rectangle around object)
  - **Centroid** (center point)

🔹 Step 7: Draw Boxes on Objects

```
rectangle(...);
plot(...);
```

- Draws **red boxes** around detected objects
- Marks center with **green +**
- Shows number of detected objects

🔹 Step 8: Extract Largest Object

```
[~, idx] = max([stats.Area]);
```

- Finds the **biggest object**

```
mask = (labeled == idx);
```

- Creates mask for that object only

```
channel(~mask) = 0;
```

- Removes everything except that object
- Final result: **only the largest object is shown**